Project Dataset

Dataset: Brazilian Olist E-Commerce Dataset

Data Integration and Data Quality

I integrated the Olist datasets using common identifiers such as customer ID, order ID, and product ID. I combined order, customer, product, payment, review, and delivery information to prepare a dataset for analysis. I checked missing values, duplicate records, inconsistent data types, and incomplete order information. I handled data quality issues where appropriate and selected relevant columns for further analysis. This helped me prepare a consistent dataset for customer segmentation, predictive modeling, and sales forecasting.

EDA and Business Findings

I performed exploratory data analysis to understand customer purchasing behavior, sales trends, product performance, payment methods, review scores, and delivery delays. I used statistical summaries and visualizations to identify patterns in the data. I examined how delivery performance and customer reviews could affect customer satisfaction. These findings helped me understand the business problems and identify areas where the company could improve its customer experience and sales performance.

Feature Engineering

I created useful features from the available order and customer information. I used RFM analysis, which includes recency, frequency, and monetary value, to understand how recently customers purchased, how frequently they ordered, and how much they spent. I also prepared delivery-related features by comparing estimated and actual delivery dates. These features helped me understand customer value, identify purchasing patterns, and prepare relevant inputs for clustering and predictive modeling.

Customer Segmentation

I applied K-Means clustering to group customers according to their purchasing behavior. Before clustering, I scaled the selected numerical features so that variables with larger values would not dominate the results. I tested different values of K from 2 to 6 and used

the Elbow Method and Silhouette Score to compare the clustering solutions. I then examined the customer count, recency, frequency, and monetary value of each cluster to understand its characteristics. These segments can help the business identify valuable customers and plan suitable retention and marketing campaigns.

Predictive Modeling and Evaluation

I developed a classification model to predict a low customer review outcome, defining the target before training. I separated the input features from the target and used appropriate preprocessing and train-test splitting to reduce data leakage. I trained and compared multiple classification models, including a baseline model, using accuracy, precision, recall, F1-score, ROC-AUC where appropriate, and confusion matrices. I compared their performance on unseen test data and selected the final candidate based on the evaluation results. The model can help the business identify orders that may be associated with poor customer feedback.

Model Improvement and Tuning

I used five-fold cross-validation and hyperparameter tuning to improve the selected model. I compared the original model with the tuned version to check whether the changes improved performance. I also compared training and testing results to identify possible overfitting or underfitting. This allowed me to select a model based on its overall performance rather than relying only on training accuracy.

Sales Time Series Analysis and Forecasting

I aggregated the available order data into a monthly sales time series and visualized the historical trend. I used a forecasting method to estimate sales for the next planning period and evaluated it using a reserved test period and an appropriate error metric, such as MAE or RMSE. The forecast can support inventory planning, staffing, and sales decisions. However, its reliability depends on historical patterns continuing and on the quality and coverage of the available data.

Data Degradation

I identified data quality issues such as missing values, incomplete records, inconsistent identifiers, and possible delivery or review data limitations. These issues can affect customer profiles, model predictions, and forecasting accuracy. I addressed relevant issues during data preparation, but some limitations may remain because the analysis depends on the information available in the original dataset.

Data Equity

I considered the possibility that the dataset may not represent all customer groups and purchasing situations equally. Missing records and differences in customer activity may influence the results of segmentation and prediction. I would recommend monitoring model performance across relevant groups, checking for systematic errors, and using human review before making important customer-related decisions.

Business Questions and Answers

1. What was the most difficult data integration problem, and how did I solve it?

The main challenge was combining multiple Olist tables that contained different types of information and were connected through different identifiers. I used the relevant customer, order, and product keys to join the datasets and checked missing values and duplicate records after integration. This helped me prepare a more consistent dataset for analysis while reducing the risk of incorrect joins and repeated records.

2. Which engineered feature added the most business value, and why?

I found RFM features useful because they describe customer purchasing behavior from three different perspectives: how recently a customer purchased, how often they purchased, and how much they spent. These features support customer segmentation and help identify customers who may deserve retention campaigns or targeted offers. The final choice of the most valuable feature should be supported by the actual cluster profiles and model results.

3. Which customer segment is the most valuable, and what evidence supports my answer?

I would identify the most valuable segment by comparing the monetary value, purchase frequency, and recency of each cluster. A segment with higher spending and repeat purchases may represent stronger customer value, while recency helps show whether

customers are still active. I would use the actual cluster summary from my notebook to name the segment and support the conclusion with its customer count and average RFM values.

4. Which predictive model performed best, and why?

I compared the classification models using accuracy, precision, recall, F1-score, ROC-AUC where appropriate, and confusion matrices. I selected the final candidate according to its test performance and suitability for predicting low customer reviews. Precision and recall were important because false predictions could lead to missed customer-service problems or unnecessary interventions. The final model name and its actual evaluation scores should be reported from my model comparison table.

5. What evidence shows that my model is not simply memorizing the training data?

I compared the model's training and test performance and used five-fold cross-validation during model improvement. Similar performance across training, testing, and validation provides evidence that the model can generalize beyond the training data. However, these checks do not guarantee perfect real-world performance, so the model should also be monitored when new data becomes available.

6. Which business patterns were surprising or unexpected?

I examined customer purchasing patterns, product performance, delivery delays, monthly sales, and review scores to identify relationships that could be useful for the business. One important area was the relationship between delivery performance and customer feedback, because late deliveries may contribute to a poor customer experience. The specific unexpected pattern should be stated using the visualizations and findings from my notebook rather than assuming that every observed relationship is causal.

7. What does the monthly forecast suggest for the next planning period?

I used the historical monthly sales series to estimate sales for the next planning period and evaluated the forecast against a reserved test period. The forecast can help the business plan inventory, staffing, and sales activities. Management should consider the forecast error and uncertainty before making decisions, particularly if demand changes or historical patterns do not continue.

8. What data limitation could most affect my conclusions?

The main limitations include missing or incomplete records, possible bias in the available customer data, and the fact that the dataset represents historical transactions rather than all future customer behavior. Review information may also be unavailable for orders that

have not yet received feedback. These limitations can affect segmentation, predictive performance, and forecasting, so the results should be interpreted in the context of the available data.

Conclusion

I used the Olist dataset to complete data integration, data quality checks, exploratory analysis, feature engineering, customer segmentation, predictive modeling, model tuning, and monthly sales forecasting. The work provides a foundation for understanding customer value, identifying potential low-review outcomes, and supporting business planning. The results can guide practical decisions, but the final recommendations should be based on the measured results and the limitations identified during the analysis.